

```
In [1]: import pandas as pd

df = pd.read_csv("/home/hanysaa/sales_clea
print(df.shape)
df.head()
```

(292, 6)

```
Out[1]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>order_id</th>
<th>date</th> <th>product</th> <th>price</th>
<th>qty</th> <th>zip</th> </thead> <tbody>
<th>0</th> <th>1</th> <th>2</th> <th>3</th>
<th>4</th> </tbody> <tbody></tbody>
```

1254	2026-04-06	mug	11.36	4	60614
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1057	2026-05-30	charger	14.79	3	30303
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1150	2026-05-06	mug	5.50	1	10001
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1066	2026-05-30	notebook	37.53	1	98101
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1114	2026-04-06	webcam	45.59	4	90405
------	------------	--------	-------	---	-------

```
In [2]: print("Price mean:", df["price"].mean())
print("Price median:", df["price"].median(
print("Price standard deviation:", df["pri

print("Qty mean:", df["qty"].mean())
print("Qty median:", df["qty"].median())
print("Qty standard deviation:", df["qty"])
```

Price mean: 55.52315068493151

Price median: 37.53

Price standard deviation: 71.76708493504132

Qty mean: 2.9863013698630136

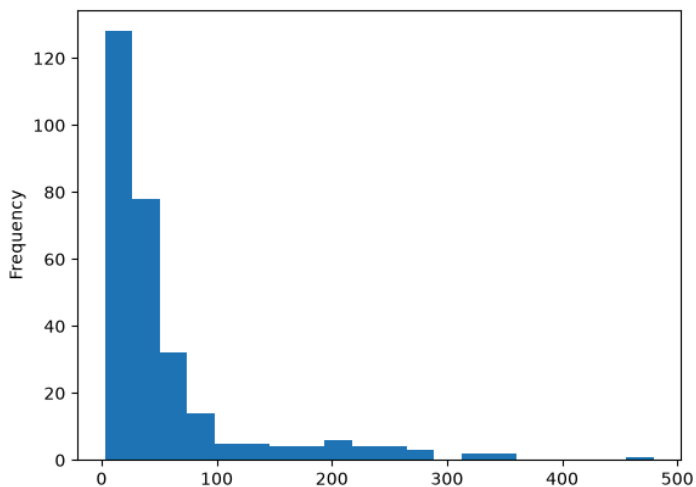
Qty median: 3.0

Qty standard deviation: 1.4092785381929205

<p>I would use the median for price,becuae the mean is way higher than the median ,probably on or more items is more excpincve pull it high which represent right-skewed.</p>

```
In [3]: df["price"].plot(kind="hist", bins=20)
```

```
Out[3]: <Axes: ylabel='Frequency'>
```



<p>the price distribution showing a few excpincve price in the left going down to the right which represent a right-skewed</p>

```
In [4]: df["revenue"] = df["price"] * df["qty"]  
df.head()
```

```
Out[4]: <style scoped> .dataframe tbody tr th:only-of-type {
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</thead> <tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

1254	2026-04-06	mug	11.36	4	60614	45.44
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1057	2026-05-30	charger	14.79	3	30303	44.37
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1150	2026-05-06	mug	5.50	1	10001	5.50
------	------------	-----	------	---	-------	------

1066	2026-05-30	notebook	37.53	1	98101	37.53
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1114	2026-04-06	webcam	45.59	4	90405	182.36
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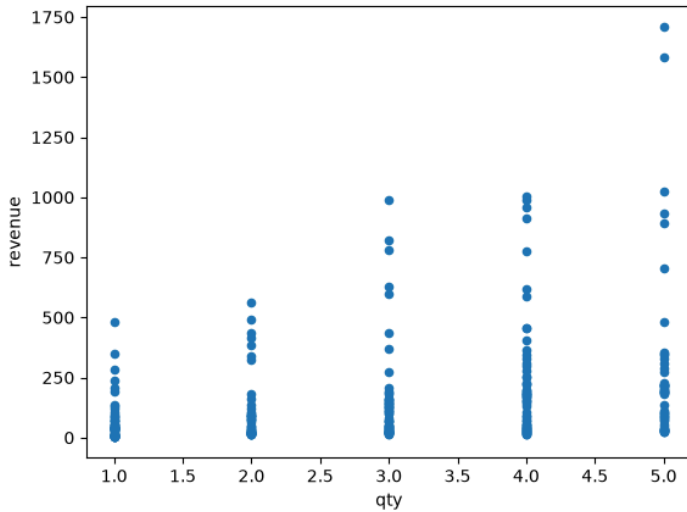
```
In [5]: df.corr(numeric_only=True)
```

```
Out[5]: <style scoped> .dataframe tbody tr th:only-of-type {
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<th>order_id</th> <th>price</th> <th>qty</th>
<th>zip</th> <th>revenue</th> </tbody> <tbody>
</tbody>
```

1.000000	-0.005070	0.031420	-0.014784	0.004181
-0.005070	1.000000	-0.041982	0.023812	0.821001
0.031420	-0.041982	1.000000	0.086869	0.312793
-0.014784	0.023812	0.086869	1.000000	0.047090
0.004181	0.821001	0.312793	0.047090	1.000000

```
In [6]: df.plot.scatter(x="qty", y="revenue")
```

```
Out[6]: <Axes: xlabel='qty', ylabel='revenue'>
```



<p>there is a postive realtion between the increase in revenue and quantaty sold ,both increase in the same way but not strong relation (0.31) , however ,there is other factor beacue the price alsoo has a stong relation (0.82) .</p>

```
In [7]: numbers = [2, 4, 4, 6, 9]

print("Mean:", sum(numbers) / len(numbers))

numbers.sort()
print("Median:", numbers[len(numbers) // 2])
```

Mean: 5.0
Median: 4

<p>after doing the mean and median manulayy..
Mean = 5 , Median = 4 whcih is matchiang with
python.</p>

```
In [ ]:
```